Native independent replay of the retained degree-83 online observation

Dated 2026-10-12. Source snapshot: `aburan28/crypto` commit `a8194d616ed8c04489ec46177d369f1b1670360a`.

The standalone C++ validator independently reproduced the archived curve, subgroup, factor-base action, three four-point witnesses and six recovered scalar answers. It used ordinary polynomial-basis GF(2^83) arithmetic and imported no research solver. This is a new archive correctness check; the IC and rho algorithms were not rerun or retimed by this validator.

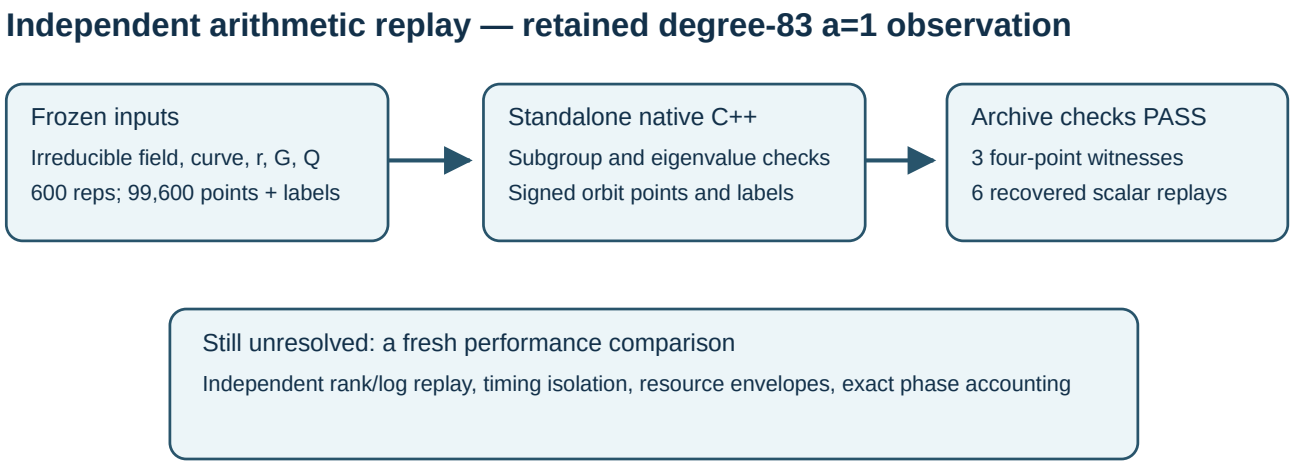


Figure 1: Scope of the independent arithmetic replay; the fresh algorithm comparison remains separate.

Exact mathematical instance

The model is `icv1-f2m83-t6151469093347-cdcc5432`: $y^2+xy=x^3+x^2+1$, modulo `x^83+x^7+x^4+x^2+1`. Independent Rabin irreducibility passed. The Koblitz trace recurrence gives `#E = 967140655691088192855606 2`, equal to subgroup prime `r = 8569786107849059` times cofactor `1128547018`. The subgroup prime was checked by the deterministic seven-base Miller–Rabin test valid over 64-bit integers.

The generator and public target have these polynomial-basis coordinates:

```
G_x = 661424465139818486029 2015
G_y = 886826041002924939153 9529
Q_x = 265589834553716618053 2960
Q_y = 388152630799167326529 2615
```

Both are nonidentity points on the curve, annihilated by r. The validation-only sidecar scalar `3141592653589793` replays to this target. No new unknown-scalar solution was computed.

The scalar `lambda = 2998004823950457` satisfies `lambda^2-lambda+2=0 (mod r)`, has exact order 83 and obeys `tau(G)=[lambda]G`. Every one of the 600 factor-base representatives lies in the subgroup. All 99,600 retained points were checked to be distinct, on curve, exactly the 166 signed Frobenius images of their representatives, with correct `(column,±lambda^k)` labels. Each orbit closes after 83 shifts. All three four-point target witnesses re-add to Q; their x codes and two pinned intermediate sum abscissae match the retained receipts.

Recomputed archived interval arithmetic

These are historical recorded milliseconds on the original contended ARM64 host, not measurements on this Linux checker. Three repeats share one public point. Ratios use internal rho recorded total / IC target interval.

Run	IC target ms	Rho internal total ms	Rho/IC	Unassigned IC ms
R1	7939.137708	64952.483042	8.181301979	0.000458

Run	IC target ms	Rho internal total ms	Rho/IC	Unassigned IC ms
R2	7746.014250	42987.579541	5.549638582	0.000084
R3	5623.275125	71253.689916	12.671208207	0.000541

Each five-phase sum matches the recorded `target_phase_sum_ms` but differs slightly from `target_ms`. The gap is less than one microsecond; an exact exclusive-phase contract must assign that elapsed bookkeeping cost rather than state exact equality. Rho totals equal setup+walk+validation; setup is 7.556, 7.816083 and 7.516250 ms. Its target dependence remains unresolved. The source README rounded rho figures differ from the retained internal totals; the CSV and this check use the latter.

Each IC repeat records 8,845,441 probes. Rho walk steps are 6,608,900, 5,958,775 and 12,179,440. Those counters count different work and do not form an algorithm speedup metric. Reusable IC construction/rank setup is excluded from this online comparison.

What remains unresolved

The 600 rank equations and recovered factor-log database are not retained in the checked summary rows; their independent algebraic replay remains unverified. The final recovered scalars and retained target witnesses pass independent arithmetic replay. This does not prove the implementation's rank construction history or absence of scalar input beyond the retained metadata/launch description.

The archived observations have no qualifying isolation receipt, enforced matched memory cap, complete canonical candidate/workload manifests or independent execution replay of both algorithms. The IC process used 12 threads for reusable rank setup; the current source defaults target extraction to one thread unless `KIC_TARGET_THREADS` is set. No target-thread envelope is frozen in the historical producer receipts.

The requested fresh same-point benchmark remains a separate obligation, being handled by the parent task. This validator does not discharge it. No repository, ledger or canonical graph was edited. The archival observation retains its exploratory status; existing scoreboard conclusions do not change from this arithmetic recheck.

Reproduction and evidence

```
g++ -std=c++17 -O2 -Wall -Wextra -Werror check.cpp -o check
CHECK_SOURCE_DIR=/workspace/repos/crypto/experiments
CHECK_SOURCE_DIR=$CHECK_SOURCE_DIR/koblitz-single-target-n83-20261006
./check "$CHECK_SOURCE_DIR" receipt.json
```

Compiler: Debian g++ 14.2.0; Linux x86-64. The run completed successfully. `source_hashes.sha256` binds the fixture, full base and all three producer and summary records. `receipt.json` contains exact arithmetic outputs. `check.cpp` is the editable independent arithmetic implementation.

Receipt SHA-256: `7d4e290524ea9b83ff75d48d20b17449fb60a6991bae787fdfee53ef28d461d5`.

The pinned experiment evidence contains `frozen/fixture.json`, `base_n83_K600.jsonl`, and the R1, R2 and R3 receipt sets under `runs/`. Exact source-file hashes are retained beside this report.

The explanatory verification diagram shows the scope of this check. Its editable SVG is included with this report.